

Electrophoretic Techniques Professional Quiz

I. Multiple Choice Questions (MCQs)

1-SDS-PAGE separates proteins primarily based on their:

- a) Shape**
- b) Charge**
- c) Molecular Weight**
- d) Isoelectric point**

Answer: (c)

2-In DNA electrophoresis, DNA molecules migrate towards the:

- a) Cathode (Negative pole)**
- b) Anode (Positive pole)**
- c) Neutral pole**
- d) Top of the gel**

Answer: (b)

3-Which of the following acts as a catalyst for polyacrylamide gel polymerization?

- a) APS**
- b) TEMED**
- c) SDS**
- d) Agarose**

Answer: (b)

4-The role of Glycerol in the sample loading buffer is to:

- a) Stain the DNA**
- b) Increase the density of the sample**
- c) Denature the proteins**
- d) Speed up the migration**

Answer: (b)

5-Ethidium Bromide is used in electrophoresis to:

- a) Increase the speed of DNA**
- b) Provide a negative charge**
- c) Visualize DNA under UV light**
- d) Maintain the pH**

Answer: (c)

6-Which gel is typically used for separating large DNA fragments?

- a) Agarose gel**
- b) Polyacrylamide gel**
- c) Starch gel**
- d) Silica gel**

Answer: (a)

7-Sodium Dodecyl Sulfate (SDS) is used to:

- a) Give proteins a positive charge**
- b) Denature proteins and give them a uniform negative charge**
- c) Act as a pH buffer**
- d) Stabilize the protein structure**

Answer: (b)

8-"Small DNA fragments move faster than large DNA fragments." This statement is:

- a) True**
- b) False**

Answer: (a)

9-The "Tracking Dye" (e.g., Bromophenol blue) is used to:

- a) Bind to the DNA permanently**

b) Monitor the progress of the run visually

c) Measure the exact size of DNA

d) Polymerize the gel

Answer: (b)

10-What happens if you increase the concentration of the gel (density)?

a) Molecules move faster

b) Molecules move slower (better separation for small fragments)

c) No change in speed

d) The gel melts

Answer: (b)

II. True or False Questions

1*DNA is naturally positively charged.

Answer: False (It is negatively charged)

2-Polyacrylamide gels are formed by the cross-linking of acrylamide and bis-acrylamide.

Answer: True

3*Small proteins are retarded more than large proteins in SDS-PAGE.

Answer: False (Large proteins are retarded more due to the sieving effect)

4-The Stacking Gel has a higher pH than the Separating Gel.

Answer: False (The Separating/Resolving gel has a higher pH)

5-Electrophoresis means "transport by electricity."

Answer: True

6-DNA Ladder is used as a reference to determine the size of unknown DNA.

Answer: True

7-Paper electrophoresis is the best method for separating large DNA molecules.

Answer: False (It is used for small charged molecules like amino acids)

8-Bubbles at the electrodes indicate that the current is flowing correctly.

Answer: True

9-SDS binds to proteins in a ratio proportional to the protein's size.

Answer: True

10-Ethidium bromide is a powerful mutagen and must be handled with care.

Answer: True

III. Fill in the Blanks (Complete)

1-..... is a detergent used to denature proteins by binding to their hydrophobic regions.

Answer: SDS (Sodium Dodecyl Sulfate)

2-The is the positive electrode that DNA fragments migrate toward during electrophoresis.

Answer: Anode

3-..... is a linear polymer extracted from seaweed, used as a medium for DNA separation.

Answer: Agarose

4-In the polyacrylamide gel system, acts as the initiator for polymerization.

Answer: APS (Ammonium Persulfate)

5-Within an agarose gel, linear DNA fragments migrate inversely proportional to the of their molecular weight.

Answer: Log₁₀

6-..... is a common tracking dye used to visualize the sample during loading and running.

Answer: Bromophenol Blue

7-DNA bands are visualized by exposing the gel to light after staining with EtBr.

Answer: UV (Ultraviolet)

8-Electrophoresis is most commonly used in clinical laboratories for isolation

and quantitation of proteins.

Answer: Serum

9-The dye used to stain protein bands after running SDS-PAGE is called

Answer: Coomassie Brilliant Blue

10-The speed of migration is affected by the and ionic strength of the buffer.

Answer: pH

IV. Short Notes

Why does DNA migrate towards the positive pole?

Answer: DNA molecules have phosphate groups that are negatively charged. In an electric field, these negative charges are attracted to the positive electrode (anode).

What is the function of the "Stacking Gel"?

Answer: Its role is to concentrate (stack) all the protein samples into a very thin, sharp line so they all enter the resolving gel at the exact same time for better resolution.

What is a "DNA Ladder"?

Answer: It is a commercial standard consisting of DNA fragments of known lengths. It is used as a "ruler" to estimate the size of unknown DNA samples in the same gel.

Mention three factors affecting the rate of electrophoresis.

Answer: 1) Net electric charge on the molecules. 2) Size and shape of the molecules. 3) Strength of the electric field applied.

How does SDS facilitate protein separation by size only?

Answer: SDS unfolds proteins (denatures them) and coats them with negative charges. This eliminates differences in shape and original charge, leaving only molecular weight as the variable for separation.

What is the "De-staining" step in protein electrophoresis?

Answer: After staining the whole gel with Coomassie Blue, de-staining with acetic acid/methanol removes the dye from the gel matrix itself while keeping

it bound to the protein bands for clear visualization.

Define Electrophoresis.

Answer: It is a technique used to separate charged molecules (like DNA, RNA, or proteins) according to their size and charge as they migrate through a medium in an electric field.

Why is Ethidium Bromide (EtBr) considered a hazardous material?

Answer: Because it is an intercalating agent that can insert itself between DNA bases, which makes it a powerful mutagen that can lead to cancer or genetic mutations.

What is the difference between Agarose and Polyacrylamide gel?

Answer: Agarose has larger pores and is used for large molecules like DNA. Polyacrylamide has much smaller, adjustable pores and is ideal for separating proteins and small DNA fragments.

What is the purpose of adding Glycerol to the Loading Buffer?

Answer: Glycerol increases the density of the sample, making it heavier than the running buffer. This ensures the sample sinks to the bottom of the well and doesn't float away.